

Giant dark matter–neutron star halo and the nature of gamma bursts

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It is shown that dark matter forms a halo around the galaxy. The size of the halo and the moment of its formation are respectively estimated to be $R_0 = 300\text{--}400$ kpc and $Z_0 = 5\text{--}7$. The same halo is formed by relic neutron stars (RNS) occurring in the protogalaxy as a result of supernova bursts which cause a change in the original chemical composition. The second component of the neutron stars, which were born in the galactic disk and were pulsars at the early stage of their evolution, occurs much later. The statistic distribution of γ -bursts determined by both neutron-star components is shown to be in good agreement with observations. A method of direct verification of the model is proposed.

1. Dark matter and neutron star halo

It is a well-known fact that in the course of galaxy formation, the primary matter consisting of hydrogen and helium is chemically transformed to become a galactic gas with a large amount of heavy elements [1]. Such a transformation in an early protogalaxy is due to a rapid contraction and burning-out of massive stars, which results in supernova flares and the formation of neutron stars [2]. These neutron stars do not undergo collisions and hence move as the dark matter which forms a giant spherical halo. The density distribution of neutron stars is similar to that of the dark matter, ρ_d , given by the relation [3,4]

$$\rho_d = Kr^{-\alpha}, \quad \alpha = 1.7\text{--}1.8. \quad (1)$$

The effective radius R_0 of the dark matter cluster can be expressed, using (1), in terms of its total mass M or the baryon matter mass M_G ,

$$R_0 = \left(\frac{(3-\alpha)M}{4\pi K} \right)^{1/(3-\alpha)} = \left(\frac{(3-\alpha)M_G}{4\pi Kp} \right)^{1/(3-\alpha)}. \quad (2)$$

The latter expression takes into account that at the moment of self-trapped cluster formation the baryon ρ_b and the dark ρ_d matters are uniformly distributed:

the parameter $p = \rho_b/\rho_d$ in (2) is a universal constant which is the same for all galaxies.

It is exceedingly important that at the moment of self-trapping and the formation of a protogalactic dark-matter cluster, the Hubble expansion inside the cluster ceases. It becomes an independent element in the universe, and the Hubble expansion only increases the distance between the clusters. With allowance for the fact that the mean dark-matter density varies by the law [5] $\rho_d = \rho_0(t_0/t)^2$, we find the moment of protogalactic cluster formation t_0 ,

$$t_0 = \frac{t}{(1+Z_0)^{3/2}}, \quad Z_0 = \frac{R_t}{R_0} - 1, \quad R_t = \left(\frac{3}{4\pi} \frac{M_G}{\rho_b} \right)^{1/3}. \quad (3)$$

Here $\rho_b = 3 \times 10^{-31}$ g/cm³ is the mean baryon matter density at the present instant of time t [5], R_t is the effective scale of a galaxy at the moment t .

The K value in (1) can be estimated using the curve of galaxy rotation. For our galaxy

$$K = 0.7 \times 10^{14} \text{ g cm}^{-1.3}, \quad \alpha = 1.7, \\ K \simeq 1.2 \times 10^{16} \text{ g cm}^{-1.2}, \quad \alpha = 1.8, \quad (4)$$

which correspond to the customarily assumed dark-

matter density near the earth [6], $\rho_d = 0.3\text{--}0.4 \text{ GeV}/\text{cm}^3$. Finally, knowing K , we estimate R_0 and Z_0 ,

$$R_0 = R_{\text{halo}} \left(\frac{3\%}{p} \frac{M_G}{3 \times 10^{44} \text{ g}} \right)^{1/(3-\alpha)}, \quad (5)$$

$$Z_0 = Z_{\text{halo}} \left(\frac{p}{3\%} \right)^{1/(3-\alpha)} \left(\frac{3 \times 10^{44} \text{ g}}{M_G} \right)^{\alpha/(9-3\alpha)} - 1, \quad (6)$$

where $R_{\text{halo}} = 310 \text{ kpc}$, $Z_{\text{halo}} = 6.8$ for $\alpha = 1.7$, and $R_{\text{halo}} = 440 \text{ kpc}$, $Z_{\text{halo}} = 4.8$ for $\alpha = 1.8$.

The relic neutron star (RNS) distribution

$$\rho_n = K_n r^{-\alpha} \quad (7)$$

is similar to the dark matter distribution (1). The parameter K_n here is determined by the portion of the matter remaining in the neutron star after the burst of a relic supernova, and since the supernova mass $M \sim (10\text{--}100)M_\odot$ and the neutron star mass $M = 1.4M_\odot$ the portion of the mass, p_1 , contained in RNS must be 2%–10% of the total galactic star matter. From this we find

$$K_n = \frac{3-\alpha}{4\pi} R_0^{\alpha-3} p_1 N_G, \quad (8)$$

where $N_G \approx 2 \times 10^{11}$ is the total number of stars in the galaxy.

2. Disk neutron stars

Thus, around the galaxy there necessarily occurs a giant spherical RNS halo with a characteristic dimension of 300–400 kpc, these stars being nonuniformly distributed in space by the law (7), (9).

We now proceed to disk neutron stars (DNS). These stars are born in the disk and leave it owing to high intrinsic velocities. Since at the early stage of their lives these stars are pulsars, we know their spatial distribution, initial velocity distribution and their birth frequency $q_p \approx 1/30 \text{ years}$ [7–9]. This permits a rather accurate determination of the stationary DNS density distribution in the galaxy. The corresponding calculations have been made in refs. [10–12]. In contrast with these papers, our distribution function over initial velocities has not been thought of as isotropic because the regular motion associated

with the general star rotation $\Omega_0(\rho)$ in the galaxy has velocities much higher than the peculiar velocities of radio pulsars. Furthermore, we have used the galaxy potential which explicitly contains the contribution of dark matter and the galactic disk matter $M_d \approx 10^{11} M_\odot$. We obtain then the following expression for the DNS density ρ_{nd} ,

$$\begin{aligned} \rho_{\text{nd}}(\rho, z) &= \frac{1}{4\pi} n_0(\rho) \frac{t_G q_p}{z_0} \\ &\times \exp\left(-\frac{z^2}{2z_0^2} - \frac{\delta z}{z_0}\right) I(z, \rho), \\ I(z, \rho) &= \int_0^\infty \exp(-\tfrac{1}{2}x^2) \\ &\times \left[\tfrac{1}{2}\pi - \arcsin\left(\frac{\delta}{x^2 + (z/z_0 + \delta)^2}\right)^{1/2} \right]^{-1} dx. \end{aligned} \quad (9)$$

Here $n_0(\rho)$ is the observed density of pulsars in the disk, t_G is the galaxy life time, the z axis is directed perpendicular to the disk, δ is a nondimensional parameter,

$$\delta(\rho) = \frac{2\pi G M_G n_0(\rho)}{\Omega_0(\rho) v_0}, \quad (10)$$

v_0 is the observed dispersion of the pulsar velocities and z_0 is the characteristic width of DNS,

$$z_0 = \frac{v_0}{\Omega_0(\rho)}. \quad (11)$$

Figure 1 illustrates equal-density DNS profiles defined from (9)–(11). We can see that the characteristic width makes up approximately 1 kpc and the DNS fill a torus-shaped region (see also ref. [13]). This is exactly what should be expected because DNS that are born in the central region of the galaxy are in a deep potential well and are unable to overcome the gravitation field of the galaxy. The total number N_2 of DNS is equal to $q_p t_G \sim 10^8$, where $t_G \approx 6 \times 10^9$ years is the galactic life time.

3. Distribution of γ -bursts

Now suppose that the sources of γ -bursts are neu-

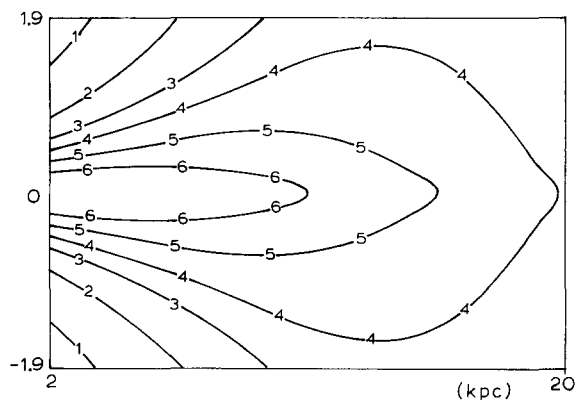


Fig. 1. Equal-density DNS profiles ρ_{nd} in $(\text{kpc})^{-3}$. $\log n = 1.75$ (1), 2.5 (2), 3.25 (3), 4 (4), 4.75 (5), 5.5 (6).

tron stars which form an extensive halo around the galaxy (cf. ref. [14]). As is known, γ -bursts are distributed over the sky very isotropically and at the same time demonstrate a considerable prevalence of near sources [15,16].

The existence of a giant RNS halo allows us not only to understand the nature of such a distribution but also to give a quantitative account of the observed statistical properties of γ -bursts. Indeed, since the total number of relic neutron stars $N_1 = p_1 N_G \sim 10^{10}$ exceeds greatly the number of disk neutron stars $N_2 \sim 10^8$, and the halo radius $R_0 \sim 400$ kpc exceeds significantly the distance between the sun and the galactic centre $r_\odot \approx 8.5$ kpc, it follows that we are in fact almost in the centre of the extensive spherical cloud whose density falls, according to (7), with increasing distance. As is shown in fig. 2, the curve $\log[N(>S)] - \log(S)$, obtained by an accurate calculation for the resultant halo and disk neutron star density, agrees with the observational data.

In fig. 3, on the p_1 - L plane for $p = 3\%$ we see the range of variation of the parameters $\langle V/V_{\max} \rangle$ and $\langle \cos \theta \rangle$ which are customarily employed in a statistical analysis of sources [17]. The threshold sensitivity $F = 10^{-7} \text{ erg/cm}^2 \text{ s}$ corresponds to the sensitivity of the COMPTON observatory [18]. We see that in the region $1\% < p_1 < 10\%$ the values $\langle V/V_{\max} \rangle$ and $\langle \cos \theta \rangle$ agree with the observations: $\langle V/V_{\max} \rangle = 0.32 \pm 0.02$ and $\langle \cos \theta \rangle = 0.04 \pm 0.03$. The quadrupole anisotropy $\langle \sin^2 b \rangle = 0.31$ – 0.32 also coincides with the experimental results. Finally, the lu-

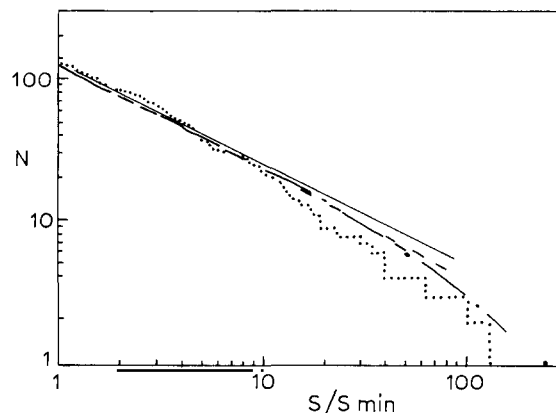


Fig. 2. The $\log[N(>S)] - \log(S)$ dependence modeled for halo and disk neutron stars. The solid line corresponds to the "standard candle" model $P(L) = \delta(L - L_0)$, the dashed line corresponds to the wide distribution function of sources over the luminosity L : $p(L) = 1/2.7L_0$, $0.3L_0 < L < 3L_0$. The dash-dotted line corresponds to the case when we do not take into account disk neutron stars. The dots indicate the observed dependence reported in ref. [18].

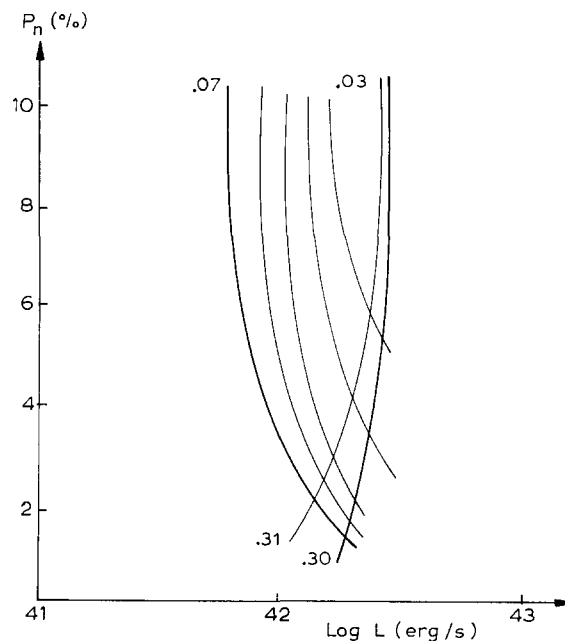


Fig. 3. Parameters $\langle V/V_{\max} \rangle$ and $\langle \cos \theta \rangle$ as functions of the luminosity L and of the portion of relic neutron stars relative to the luminescent matter p_1 for $p = 3\%$.

minosity L necessary for explaining the observed isotropy proves to be within the narrow range $8 \times 10^{41} < L < 3 \times 10^{42}$ erg/s and, accordingly, the total energy release $Q \sim 10^{42-43}$ erg.

It is noteworthy that to estimate the values of $\langle V/V_{\max} \rangle$ and $\langle \cos \theta \rangle$ we can, using the spherical distribution of the RNS halo (1), (7) and neglecting the disk stars, obtain the following simple expressions,

$$\langle V/V_{\max} \rangle = \frac{3-\alpha}{6-\alpha} \approx 0.31, \quad (12)$$

$$\langle \cos \theta \rangle = 1.0 \frac{r_{\odot}}{R_{\max}} \approx 0.04 (L/2 \times 10^{42} \text{ erg/s})^{-1/2}, \quad (13)$$

where

$$R_{\max}(L) = (L/4\pi F)^{1/2}. \quad (14)$$

Expressions (12) and (13) for $\alpha = 1.7$ agree closely with the results of the calculations shown in fig. 3.

Finally, knowing the observed frequency of the γ -bursts [18], $\nu_{\text{obs}} \approx 800 \text{ years}^{-1}$, we can estimate the burst recurrence for each neutron star, $\nu = \nu_{\text{obs}}/N_1 \sim 4 \times 10^{-8} \text{ years}^{-1}$. Consequently, during its life time $t_g \sim 10$ billion years, each halo neutron star must burst $n = \nu t_g \sim 10^3$ times so that the total energy release due to radiation in the gamma-ray range much be $Q_{\text{tot}} = nQ \sim 10^{45-46}$ erg. Such a high value restricts significantly the γ -burst generation mechanisms.

It should be noted that the dependence of the density $\rho_n(r)$ on the galaxy scales essentially affects the $\langle \cos \theta \rangle$ and $\langle \sin^2 b \rangle$ dependence on $S_{\min}$. If we suppose for example that the density of the RNS does not increase at the distance $r < r_G \approx 20$ kpc, the angular anisotropy in this case, as shown in fig. 4, will increase not so quickly. Moreover one can suppose that γ -bursts are due to the RNS only. With this proposal we get the best fit of the observed dependence $\log N - \log S$ not only in the low flux band $F \approx 10^{-7} - 10^{-6} \text{ erg/cm}^2 \text{ s}$ (GRO) but in the high flux band $F \geq 10^{-6} - 10^{-5} \text{ erg/cm}^2 \text{ s}$ (PVO) too. As we can see in fig. 4, in this case, the increasing of anisotropy $\langle \cos \theta \rangle$ is smoother also.

Finally, it is necessary to note that we considered above dark matter and neutron star halos only for the isolated galaxy. In a real situation our galaxy is connected at least with the M 31 galaxy which is at

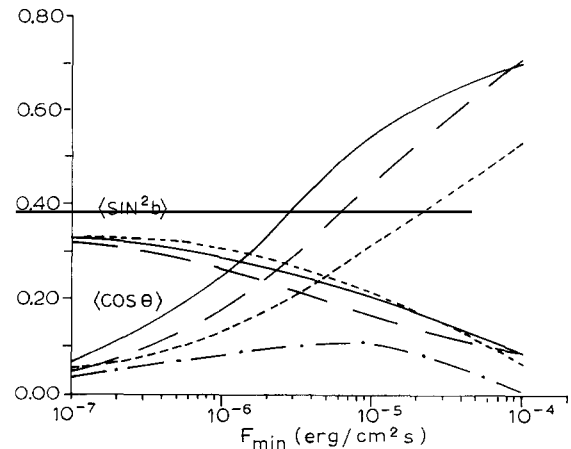


Fig. 4. Angular anisotropy $\langle \cos \theta \rangle$ and $\langle \sin^2 b \rangle$ versus the sensitivity S . The solid line corresponds to the "standard candle" case, the dashed line is the wide distribution. The dash-dotted line corresponds to the case when we take into account the RNS only.

a distance of ~ 700 kpc. That is why the halo of the RNS could be deformed at the distances $r \geq 200-300$ kpc. This halo deformation could be displayed in observations of γ -burst anisotropy. The neutron stars which belong to the M 31 galaxy could not be seen at the modern level of observational sensitivity F as the distance to them is larger than $R_{\max}$ (14). The anisotropy which Magellanic Clouds can produce in the burst distribution could be identified if the statistical material would be sufficient.

Thus we can conclude that the increase of anisotropy $\langle \cos \theta \rangle$ and $\langle \sin^2 b \rangle$ shown in fig. 4, and possibly additional investigation of peculiarities in some special directions (the M 31 galaxy, Large Magellanic Cloud) are quite sufficient tests to check the theory.

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References

- [1] A. Maeder, *Astron. Astrophys.* (1993), in press.

- [2] S.L. Shapiro and S.A. Teukolsky, *Black holes, white dwarfs, and neutron stars* (Wiley, New York, 1983).
- [3] A.V. Gurevich and K.P. Zybin, *Sov. Phys. JETP* 67 (1988) 1.
- [4] A.V. Gurevich and K.P. Zybin, *Sov. Phys. JETP* 70 (1990) 10.
- [5] Ya.B. Zeldovich and I.D. Novikov, *The structure and evolution of the universe* (Chicago Univ. Press, Chicago, 1983).
- [6] R.A. Flores, *Phys. Lett. B* 215 (1988) 73.
- [7] R.N. Manchester and J.H. Taylor, *Pulsars* (Freeman, San Francisco, 1977).
- [8] A.G. Lyne, R.N. Manchester and J.H. Taylor, *Mon. Not. R. Astron. Soc.* 213 (1985) 613.
- [9] A.G. Lyne and F.G. Graham-Smith, *Pulsar astronomy* (Cambridge, Univ. Press, Cambridge, 1990).
- [10] B. Paczynski, *Astrophys. J.* 348 (1990) 485.
- [11] D. Hartman, R.I. Epstein and S.E. Woosley, *Astrophys. J.* 348 (1990) 625.
- [12] Z. Frei, X. Huang and B. Paczynski, *Astrophys. J.* 384 (1992) 105.
- [13] M.E. Prokhorov and K.A. Postnov, *Astron. Astrophys. Trans.* (1993), in press.
- [14] I.S. Shklovskii and I.G. Mitrofanov, *Mon. Not. R. Astron. Soc.* 212 (1985) 545.
- [15] J.C. Highton and R.E. Lingenfelter, *Annu. Rev. Astron. Astrophys.* 28 (1990) 401.
- [16] A.K. Harding, *Phys. Rep.* 206 (1991) 327.
- [17] J.C. Highton and M. Schmidt, *Astrophys. J.* 355 (1990) 13.
- [18] C.A. Meegan et al. *Nature* 355 (1992) 143.